

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 3

For the tutorial in Week 4 on 8 February, let us discuss

- Ex. 6.4.2, 6.4.5, 6.4.13, 6.4.16, 6.4.20, 6.4.21, Additional Exercises, 6.5.7, 6.5.9.

Ex. 6.4.2 and Ex. 6.4.5.

- (a) Find the coefficient of x^5y^8 in $(x+y)^{13}$.
- (b) Give a formula for the coefficient of x^k in the expansion of $(x + \frac{1}{x})^{100}$, where k is an integer.

Ex. 6.4.13. Prove that if n and k are integers with $1 \leq k \leq n$, then $k\binom{n}{k} = n\binom{n-1}{k-1}$

- (a) using a combinatorial proof. [Hint: Show that the two sides of the identity count the number of ways to select a subset with k elements from a set with n elements and then an element of this subset.]
- (b) using an algebraic proof based on the formula for $\binom{n}{r}$.

Ex. 6.4.16. Show that if p is a prime and k is an integer such that $1 \leq k \leq p-1$, then p divides $\binom{p}{k}$.

Ex. 6.4.20. Show that if n is a positive integer, then $C(2n, 2) = 2C(n, 2) + n^2$

- (a) by algebraic manipulation;
- (b) using a combinatorial argument.

Ex. 6.4.21. Give a combinatorial proof that $\sum_{k=1}^n k\binom{n}{k} = n2^{n-1}$.

[Hint: Count in two ways the number of ways to select a committee and to then select a leader of the committee.]

Additional Exercise 1. How many seating arrangements are there with six boys and six girls if the table is triangular with 4 seats on each side? Two arrangements are identical if they differ by a turn of the table that keeps the chairs in place.

Additional Exercise 2. How many seating arrangements are there with six boys and six girls if the table is hexagonal with 2 seats on each side, out of which one must be occupied by a boy and the other one by a girl?

Ex. 6.5.7, and Ex. 6.5.9.

- (a) How many ways are there to choose eight coins from a piggy bank containing 100 identical pennies and 80 identical nickels ?
- (b) A book publisher has 3000 copies of a discrete mathematics book. How many ways are there to store these books in their three warehouse if the copies of the book are indistinguishable, and each warehouse should store at least one book ?